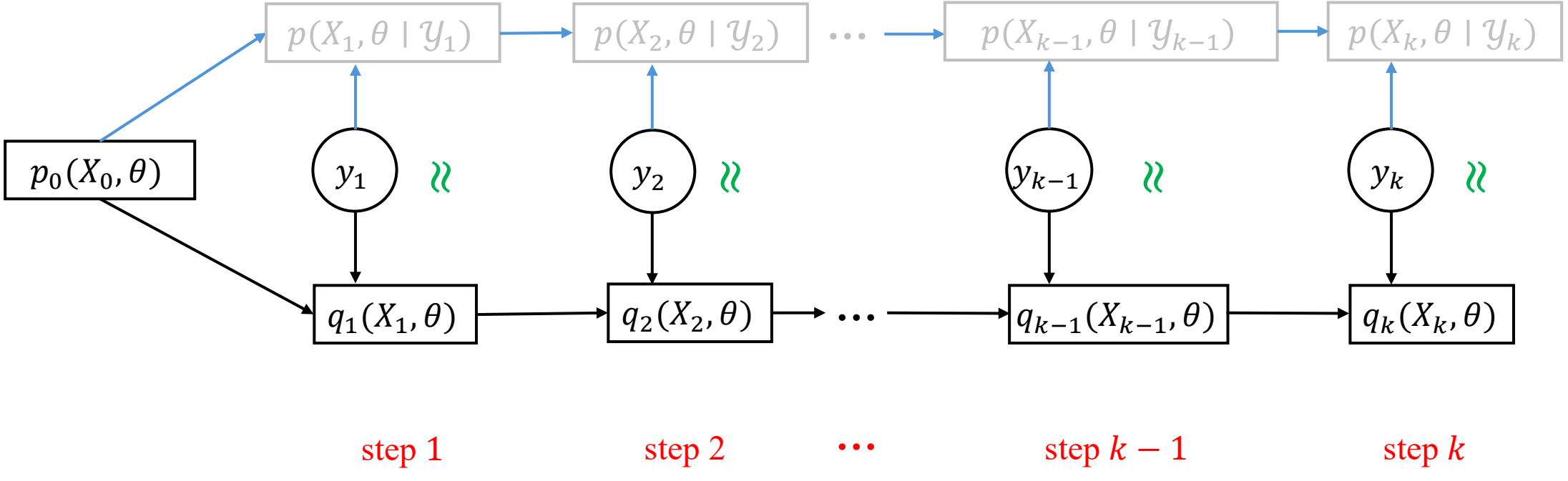

$p(X_1, \theta \mid y_1)$
$p(X_2, \theta \mid y_2)$
$p(X_{k-1}, \theta \mid y_{k-1})$
$p(X_k, \theta \mid y_k)$
$p_0(X_0, \theta)$
y_1
y_2
y_{k-1}
y_k
$q_1(X_1, \theta)$
$q_2(X_2, \theta)$
$q_{k-1}(X_{k-1}, \theta)$
$q_k(X_k, \theta)$
step 1
step 2
step $k-1$
step k